

Surgery planning

A hospital needs to define a planning for the surgeries of n patients for the next week in one of its operating rooms (OR) available. Each patient requires a specific type of surgery that can only be performed by a surgeon with the corresponding specialty. On each day, the OR is dedicated (and set up) for a single specialty for 8 hours. Using the OR for one day induces a fixed cost that depends on the specialty and the day of the week. The goal is then to assign each patient to a given day where the OR is prepared the right specialty at minimal cost.

The duration of each surgery is uncertain and may be longer than its expected value. The additional duration are distributed according to a uniform distribution, whose parameters and maximum value depend on the specialty needed by the patient. As a consequence, the total time allocated to one block may not be sufficient to perform the set of surgeries planned within it. In that case, the manager plans in advance any overtime for a given block. This enables the hospital to signal to the surgery team assigned to this block that they are likely to work longer. For legal reasons, any overtime is limited to 2h ; in addition it induces additional costs, propotional to the overtime planned, that depend on the speciality involved.

All costs depend on the day of the week : Both fixed costs and overtime costs are multiplied by 1.25 on Saturday and by 1.8 on Sunday.